

Série de Travaux Pratiques n°1 : Création de règles et de requêtes en Prolog

Exercice n°1 :

1. ? animal(mamal,X,_,_).
2. ?- animal (X, mammifere, insectivore, _, _, _, _).
3. ?- animal(X, oiseau, Y, _, _, _, _), Y\=='carnivore'.
4. ?- animal(mamal,X,_,rayures).
5. ?- animal(X, _, _, _, _, P, _), P=<1, P>10.

6-7-8

```
17 ?- animal(X,mammifere,herbivore,_,_,_,A) , animal(zebre,_,_,_,_,Z) , A<Z.  
false.  
  
18 ?- animal(X,mammifere,_,_,L,_,_) , animal(Y,reptile,_,_,L,_,_) .  
false.  
  
19 ?- animal(X,oiseau,_,Z,_,_,_) , animal(Y,oiseau,_,Z,_,_,_) , X\==Y.  
false.
```

Exercice n°2 :

1. ?enfant(bob,X).masculin(X).
2. fils(X,Y) :- masculin(X), enfant(X,Y).
3. fille(X,Y) :- feminin(X), enfant(X,Y).
4. frere_ou_soeur(X,Y) :- enfant(X,Z), enfant(Y,Z).
5. mere(X,Y) :- enfant(Y,X),feminin(X).

grand_pere(X,Y) :- pere(X,Z),enfant(Y,Z).

frere(X,Y) :- enfant(X,Z),enfant(Y,Z).

tante(X,Y) :- feminin(X),enfant(X,Z),enfant(P,Z),enfant(Y,P).

cousin(X,Y) :-enfant(X,Z1),enfant(Y,Z2), frere_ou_soeur(Z1,Z2).

Série de travaux pratiques n°2 : Prolog et les bases de données

1. `pjOran(Ref,Nat) :- projet(Ref,Nat,oran).`
2. `fpDistincts(F,P) :- fournisseur(F,_,_,VF), piece_det(P,_,_,_,VP), VF\==VP.`
3. `pjF1P1(PJ,V) :- (livraison(f1,_,PJ,_) ; livraison(_,p1,PJ,_)), projet(PJ,_,V).`
4. `?- projet(PJ,N,annaba).`
5. `?- livraison(F,_,pj1,_,_).`
6. `?- livraison(F,P,PJ,Q), Q >= 300, Q <= 750.`
7. `Jointure1 (F,P,PJ) :- livraison (F,P,PJ ,_), fournisseur (F,_,_,V1), piece_det (P,_,_,_,V2), projet (PJ ,_,V3), V1 == V2 ,V2 == V3 .`
8. `Jointure2 (P) :- livraison (F,P,_,_), fournisseur (F,_,_, annaba).`
9. `Jointure3 (P) :- livraison (F,P,PJ ,_), fournisseur (F,_,_, annaba ), projet (PJ ,_, annaba).`

Série de Travaux Pratiques n°3 : Prolog : Arithmétique, Contrôle et Négation

Q1.

$\text{dist}(xA,yA,xB,yB,D) :- D \text{ is } \text{sqrt}((xB-xA)**2 +(yB-yA)**2)$

Q2.

$\text{abs}(X,X) :- x \geq 0, !.$

$\text{abs}(X,Y) :- Y \text{ is } -X.$

Q3.

$\text{max2}(X,Y,X) :- X \geq Y, !.$

$\text{max2}(X,Y,Y).$

Q4.

$\text{max3}(X,Y,Z,M) :- \text{max2}(X,Y,MM), \text{max2}(MM,Z,M).$

Q5.

$\text{pair}(X) :- \text{mod}(X,2)=2, !.$

Série de Travaux Pratiques n°4 : La Récursivité et les Listes

Exercice n°1 :

a. Le factoriel

factoriel(0,1).

factoriel(N, Fact) :- M is N - 1, factoriel(M, M_Fact), Fact is N * M_Fact.

b. Fibonacci

fib(0,0).

fib(1,1).

fib(X,Y) :-
 X > 1 ,
 X2 is X - 2 , fib(X2, Y2),
 X1 is X - 1 , fib(X1, Y1),
 Y is Y1 + Y2.

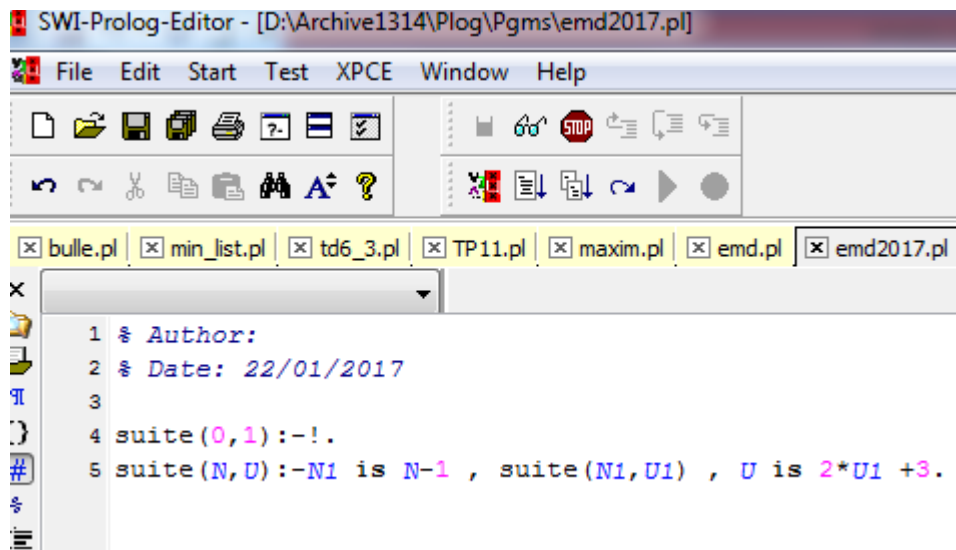
c. Le PGDC

pgcd(X,X,X).

pgcd(X,Y,D) :- X < Y, Y1 is Y - X, pgcd(X,Y1,D).

pgcd(X,Y,D) :- Y < X, pgcd(Y,X,D).

Exercice n°2.



The screenshot shows the SWI-Prolog-Editor interface. The title bar reads "SWI-Prolog-Editor - [D:\Archive1314\Plog\Pgms\emd2017.pl]". The menu bar includes "File", "Edit", "Start", "Test", "XPCE", "Window", and "Help". Below the menu is a toolbar with various icons for file operations, editing, and execution. The window title bar shows several open files: "bulle.pl", "min_list.pl", "td6_3.pl", "TP11.pl", "maxim.pl", "emd.pl", and "emd2017.pl". The main editor area displays the following Prolog code:

```
1 % Author:
2 % Date: 22/01/2017
3
4 suite(0,1):-!.
5 suite(N,U):-N1 is N-1 , suite(N1,U1) , U is 2*U1 +3.
```

Un exemple d'interrogation

```
10 ?- consult('D:/Archive1314/Plog/Pgms/emd2017.pl').
% D:/Archive1314/Plog/Pgms/emd2017.pl compiled 0.00 sec, 1 clauses
true.

11 ?- suite(6,U).
U = 253.
```

Exercice n°2 :

Sans le cut

element(X,[X|_]).

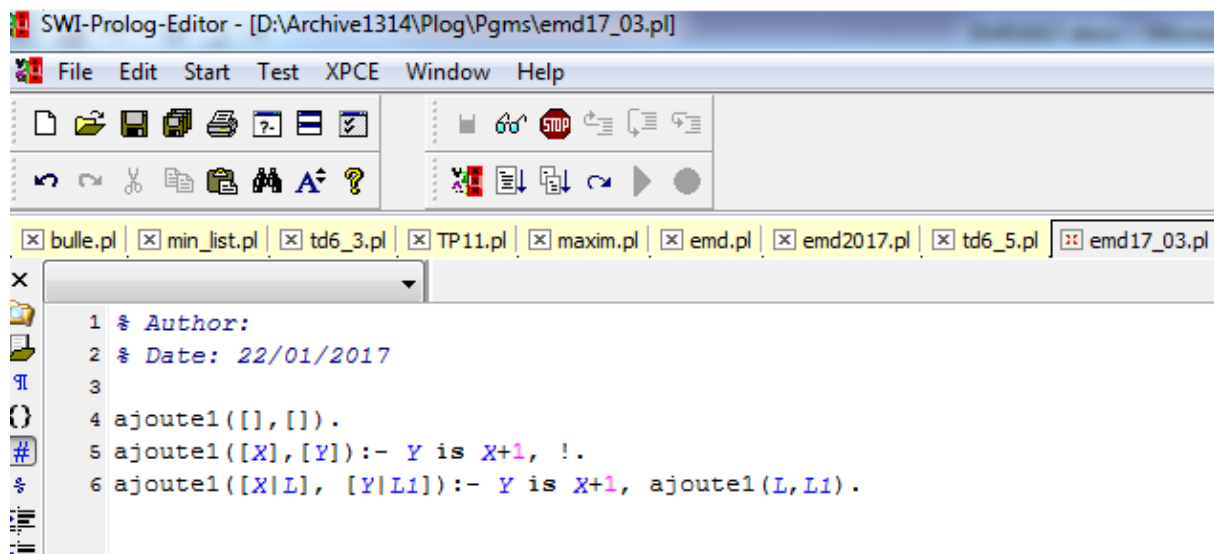
element(X,[_|L]) :- element(X,L).

Avec le cut

element(X,[X|_]) :- !.

element(X,[_|L]) :- element(X,L).

Exercice n°3 :



Exercice n°4 :

a.

`max([X], X).`

`max([X|L], X) :- max(L, MaxL), X > MaxL, !.`

`max([_|L], M) :- max(L, M).`

b.

`% my_reverse(L1,L2) :- L2 is the list obtained from L1 by reversing the order of the elements.`

`my_reverse(L1,L2) :- my_rev(L1,L2,[]).`

`my_rev([],L2,L2) :- !.`

`my_rev([X|Xs],L2,Acc) :- my_rev(Xs,L2,[X|Acc]).`

c.

```
2 compressor([], []).
3 compressor([X], [X]).
4 compressor([X,X|L1], L2) :- compressor([X|L1], L2).
5 compressor([X,Y|L3], [X|L2]) :- X\=Y, compressor([Y|L3], L2).
6
```

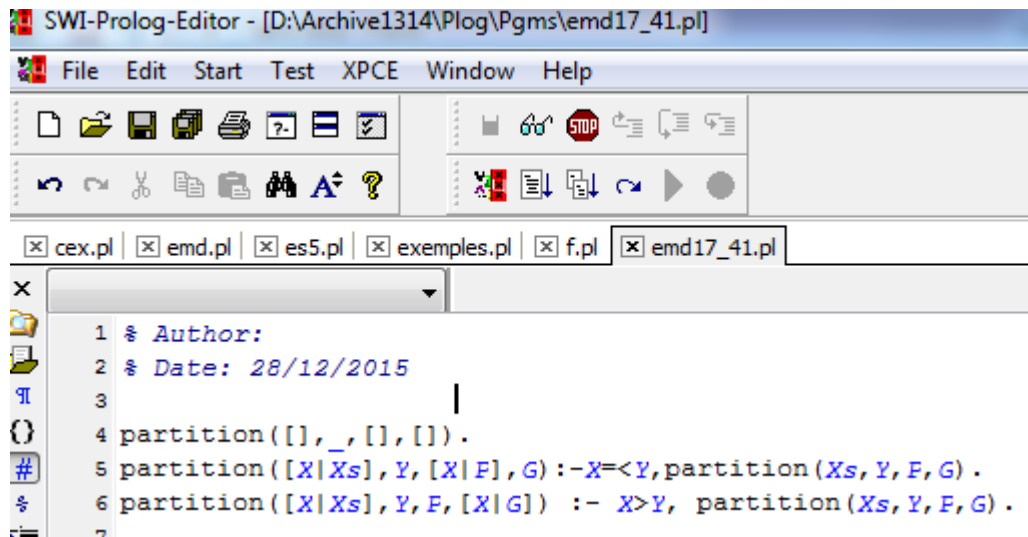
```
?- compressor([g,j,k,k,k,l,m,p,p,o], L).
L = [g, j, k, l, m, p, o] ;
?-
```

d.

```
3
4 split(L, 0, [], L).
5 split([X|L1], N, [X|L2], L3) :- N > 0, N1 is N - 1, split(L1, N1, L2, L3).
6
```

```
3 ?- split([t,h,f,k,l,a,i,p],4,L1,L2).
L1 = [t, h, f, k],
L2 = [l, a, i, p] ;
false.
```

Exercice n°5 :



```
[debug] 30 ?- partition([2,15,5,6,8,1,14,9],6,L1,L2).
L1 = [2, 5, 6, 1],
L2 = [15, 8, 14, 9] ;
```